



Storage Area Network

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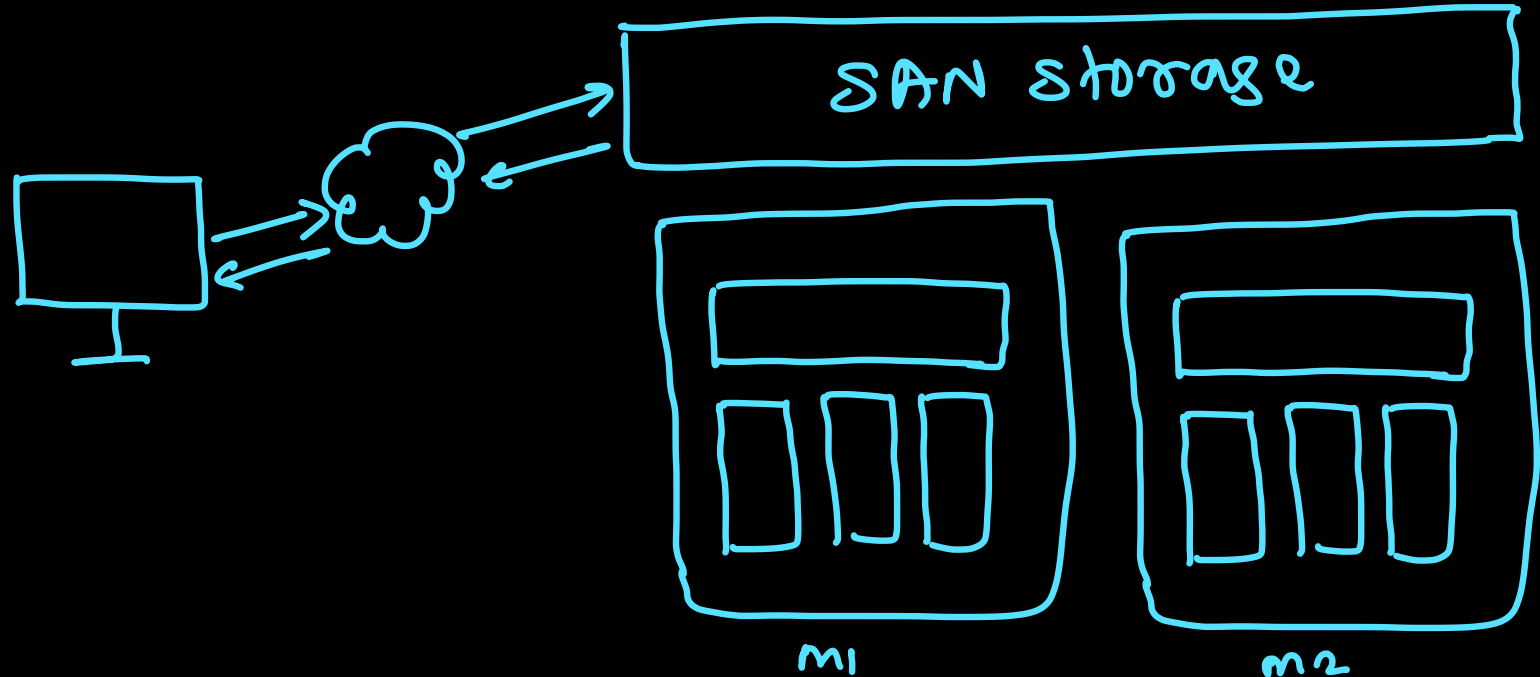


- A Storage Area Network (SAN) is a specialized, high-speed computer network that provides access to consolidated, block-level data storage RAID
- SANs are primarily used to access storage devices (such as disk arrays and tape libraries) from servers so that the devices appear to the operating system as direct-attached storage
- To a server connected to a SAN, a remote storage device located elsewhere on the network appears and behaves exactly as if it were a physical hard drive plugged directly into the server

Block based storage

File based storage

Object based storage



SAN vs. NAS

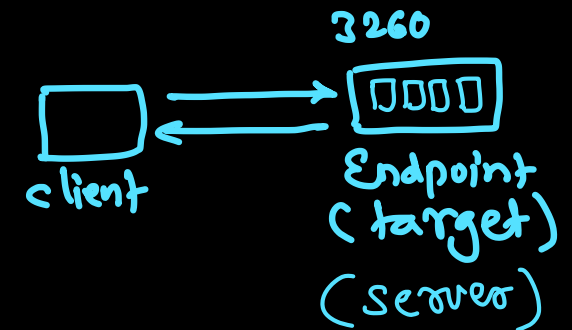


Feature	<u>SAN (Storage Area Network)</u>	<u>NAS (Network Attached Storage)</u>
<u>Primary Access Level</u>	<u>Block-level</u>	<u>File-level</u>
<u>How OS Sees It</u>	<u>Appears as a local, unformatted hard drive</u>	<u>Appears as a network file share (folder)</u>
<u>Protocols</u>	<u>iSCSI, Fibre Channel (FC), NVMe-oF</u>	<u>SMB/CIFS, NFS, AFP</u>
<u>Typicated Usage</u>	<u>Databases, virtualization (VMware/Hyper-V), high-performance applications</u>	<u>File sharing, document storage, backups, media streaming</u>
<u>Granularity Control</u>	<u>OS handles file system (formatting, partitioning)</u>	<u>NAS handles file system internally</u>
<u>Typical Network</u>	<u>Dedicated high-speed network (often isolated)</u>	<u>Shared Local Area Network (LAN)</u>

How SAN Works: The iSCSI Protocol



- While Fibre Channel SANs use specialized hardware, the most common approach for SMBs and homelabs is iSCSI (Internet Small Computer System Interface)
- SCSI encapsulates SCSI commands inside TCP/IP packets, allowing block storage to be transmitted over standard Ethernet networks
- **Key iSCSI Components**
 - Initiator: The client (server, Hypervisor) that connects to the SAN
 - Target: The endpoint on the storage server that receives connections
 - LUN (Logical Unit Number): A unique identifier for a block device within a target
 - Portal: The IP address and port (default 3260) where the target listens
 - Extent: The backend storage resource (often a ZVOL) being shared



Building a SAN: The Core Components



- **Storage Operating System** → *customized Linux system*
 - Platforms like TrueNAS (CORE or SCALE) provide the software intelligence.
 - TrueNAS uses the ZFS filesystem which offers:
 - High data integrity and checksumming
 - Snapshots and clones
 - Compression and deduplication
 - Advanced caching (L2ARC, SLOG)
- **Network Infrastructure**
 - **Dedicated Network:** Best practice is isolating SAN traffic on separate physical or VLAN interfaces to prevent congestion
 - **High Bandwidth:** 10GbE (or higher) is strongly recommended
 - **Jumbo Frames:** Setting MTU to 9000 can improve throughput for large sequential transfers
 - **Multipathing:** Configuring multiple network connections improves both performance (load balancing) and fault tolerance
- **Storage Hardware**
 - **For virtualization (high IOPS):** All-flash configurations (NVMe or SATA SSDs)
 - **For capacity (backup/archiving):** High-capacity HDDs in RAID/ZFS pools

SAN Use Cases



- Virtualization (VMware, Hyper-V, Proxmox)
 - VMs require block-level access; SAN provides shared datastores with VMFS locking
- Databases (SQL, Oracle, PostgreSQL)
 - Direct block access reduces latency and improves transaction performance
- High Availability Clusters
 - Multiple servers can access the same LUN with proper cluster coordination
- Boot-from-SAN
 - Diskless servers boot entirely from network storage
- Backup Targets
 - High-performance block storage for backup applications

Pros of SAN



■ Performance

- High throughput & low latency
- Dedicated network (Fibre Channel or high-speed Ethernet) eliminates contention. Block-level access bypasses file system overhead on the storage server.

■ Scalability

- Massive capacity expansion
- You can add shelves of drives (JBODs) or entire new arrays without taking servers offline. Many SANs scale to petabytes.

■ High Availability

- Redundant paths & failover
- Multipathing (MPIO) ensures that if one network switch, cable, or port fails, I/O continues seamlessly over another path.

■ Boot-from-SAN

- Diskless servers
- Servers can boot entirely from the SAN, allowing rapid replacement of failed servers without reimaging drives.

■ Centralized Management

- Single pane of glass
- All storage (for dozens or hundreds of servers) is provisioned, monitored, and backed up from one interface rather than managing local disks per server.

■ Advanced Features

- Snapshots, cloning, replication
- Enterprise SANs (and ZFS-based systems like TrueNAS) offer instant snapshots for backup, fast cloning for testing, and async/sync replication to remote sites.

■ VMware/Cloud Integration

- Shared datastores
- Multiple hypervisor hosts can simultaneously read/write the same LUN (with cluster filesystem like VMFS), enabling vMotion, DRS, and HA clustering.

Cons of SAN



■ Cost

- Expensive to acquire and operate
- Fibre Channel switches, HBAs (Host Bus Adapters), enterprise disk arrays, and the required 10/25/40GbE networking are significantly more expensive than DAS or basic NAS.

■ Complexity

- Steep learning curve
- Requires specialized knowledge of zoning, LUN masking, multipathing, WWNs (World Wide Names), initiator/target configuration, and often storage networking separate from LAN administration.

■ LUN Management Overhead

- Capacity planning is critical
- You must allocate LUN sizes in advance. Shrinking a LUN is often difficult or impossible. Growing a LUN may require rescanning and filesystem expansion on every connected server.

■ Single Failure Domain (if misconfigured)

- Misconfiguration affects many servers
- A mistake in LUN masking or zoning can accidentally expose a LUN to the wrong server, corrupting data. A storage controller failure affects all connected servers.

■ Not Ideal for Small Deployments

- Overkill for 1-3 servers
- For a single server or small cluster, DAS or a simple NAS (with iSCSI) often provides sufficient performance at a fraction of cost and complexity.

■ Block-Level Locking Limitations

- Clustering requires cluster-aware filesystems
- You cannot simply connect two servers to the same LUN without a cluster filesystem (VMFS, OCFS2, GFS2) or volume manager. Otherwise, data corruption occurs.

■ Network Dependency

- Performance suffers on shared Ethernet
- Running iSCSI over a congested LAN (without dedicated VLAN or physical interfaces) leads to timeouts, disconnections, and application errors.

When to Choose SAN vs. Alternatives



■ Single home server / homelab

- DAS or NAS
- SAN complexity provides no benefit

■ Small business (2-5 servers)

- NAS with iSCSI
- TrueNAS or similar gives SAN-like features without dedicated hardware

■ Medium business with virtualization

- SAN (iSCSI or FC)
- Multiple hypervisors need shared block storage for vMotion/HA

■ Enterprise with databases

- SAN (FC or NVMe-oF)
- Requires low latency and high IOPS at scale

■ Backup target only

- NAS or DAS
- File-level access is sufficient; block storage adds complexity

■ Edge/IoT or remote office

- Cloud storage or DAS
- SAN requires skilled on-site management

TrueNAS → FreeNAS



- TrueNAS is an open-source, unified storage operating system developed by iXsystems (now doing business as TrueNAS)
- It allows you to turn standard x86-64 hardware into a powerful Network Attached Storage (NAS) or Storage Area Network (SAN) platform .
- Originally named FreeNAS (first released in 2005), the project has evolved significantly
- In 2020, iXsystems introduced a Linux-based variant called TrueNAS SCALE, and in 2025, the FreeBSD-based TrueNAS CORE and Linux-based SCALE lines were unified into a single, community-driven product simply called TrueNAS Community Edition
- The current stable release series as of late 2025 is 25.10 "Goldeye"

Advanced Features and Capabilities



■ Virtualization and Apps

- TrueNAS is not limited to storage
- You can run applications and virtual machines directly on the platform
- **Applications (Apps):** Built-in support for **Docker containers**. You can deploy popular apps like **Plex**, **Nextcloud**, **Prometheus** from community or official catalogs
- **Virtual Machines (VMs):** TrueNAS includes a **KVM** (Kernel-based Virtual Machine) hypervisor, allowing you to run full operating systems like Windows, Linux, or FreeBSD alongside your storage services

■ High Availability (HA) and Replication

- **High Availability (Enterprise):** For mission-critical environments, two TrueNAS controllers can be configured in an active-passive HA cluster. If the primary controller fails, the secondary takes over with minimal downtime
- **Replication:** You can automatically replicate ZFS snapshots to another TrueNAS system (local or remote), providing an efficient disaster recovery solution

■ Centralized Management: TrueCommand

- For managing multiple TrueNAS systems (clusters or fleets), iXsystems provides **TrueCommand**
- This is a centralized web-based tool that gives you a single pane of glass for monitoring, alerting, and administration across all your storage appliance